

Abdiel M. Vallejo Padró

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Website →



PROFESSIONAL SUMMARY

Computer Science and Engineering student at UPRM with experience in autonomous systems, software development, AI, aerospace simulation, and full-stack web development. Experienced in autonomy, fault-tolerant architectures, sensor integration, and data-driven engineering tools.

EDUCATION

University of Puerto Rico, Mayagüez (UPRM)

Mayagüez, PR

B.S. Computer Science and Engineering, Project Management Minor | GPA: 3.5

Expected 2029

TECHNICAL EXPERIENCE

NASA RASC-AL Competition | Lead Hardware & Autonomous Systems Engineer

Jun. 2025

- Designed a **fault-tolerant autonomous computing architecture** using dual NASA HPSC processors and an AMD Versal AI Core, incorporating SHERPA-based recovery logic to support resilient mission operations in a lunar environment.
- Developed an **autonomous ConOps workflow** integrating health monitoring, AI-driven task prioritization, path planning, fault handling, and mission execution logic for rover operations.
- Validated sensor placement and autonomous perception coverage through **ROS Noetic and Gazebo** simulations on Ubuntu, supporting navigation and system design decisions.

SPARC, UPRM | Research Member

Aug. 2024 – Present

- Conducted **AI and autonomy trade studies** for lunar rover systems, evaluating technical approaches and translating findings into system-level recommendations for team development.

Solar Boat, UPRM | Computer Engineering Division

Aug. 2026 – Present

- Developing and maintaining the team's website using HTML and CSS, leveraging **Git and GitHub** for version control, collaboration, and code management.

EXTRACURRICULAR PROJECTS

Lockheed Martin AI/ML Hackathon | PyTorch, AWS SageMaker, Team of 3

Sep. 2026

- Fine-tuned a COCO-pretrained Faster R-CNN ResNet-50 FPN v2 on 827 KITTI-labeled recycling-plant images using PyTorch and AWS SageMaker, achieving **0.817 VOC mAP@50** on the organizers' held-out test set.
- Identified near-duplicates affecting **61%** of the official validation set and built a leak-resistant temporal split with duplicate purging and multi-seed experiments, improving clean-validation mAP@50 from **0.846 to 0.874**.

Lilly AI Mini Hackathon | Team of 5

Sep. 2026

- Conceptualized **PayFlow AI**, an AI-driven payroll request management solution, contributing to UX design and workflow development to improve request tracking and user experience.

AI Winter School | University of Notre Dame

Jan. 2025

- Built and analyzed **linear regression models** using Python, Jupyter Notebook, and Google Colab during an intensive AI program, applying data analysis concepts to hands-on modeling exercises.

PERSONAL PROJECTS

Barber Bot | WhatsApp Booking Automation

2025 – Present

- Developed a **conversational booking assistant** using Python, FastAPI, Groq, Redis, and the Meta WhatsApp API to automate appointment scheduling, reminders, cancellations, and customer interactions.
- Integrated **Google Calendar** into the booking workflow to provide real-time availability and automate scheduling coordination through WhatsApp.

Stock Analyzer | Open Source | Financial Analysis Platform

2026

- Built an **11-module financial analysis backend** with Python and FastAPI, exposing **13 API endpoints** for DCF, VWAP, Sharpe ratio, VaR, and SEC EDGAR insider analysis.
- Integrated **yfinance, Alpha Vantage, and SEC EDGAR** with caching at zero API cost, enabling scalable access to financial data and customizable market sentiment analysis.

TECHNICAL SKILLS

Programming & Software : Python, C++, C, SQL, JavaScript, FastAPI, React, Git, GitHub

AI, Data & Simulation : Pandas, NumPy, Power BI, ROS Noetic, Gazebo, MATLAB, Jupyter Notebook, PyTorch, AWS SageMaker

Engineering Tools : SolidWorks, Redis, Ubuntu